

# DRISHTIX v2.0 TACTICAL

## Intelligent Border Surveillance & Tactical Command Center Platform — Operations Dossier

|                                           |                                          |                                      |
|-------------------------------------------|------------------------------------------|--------------------------------------|
| Project: DRISHTIX MVP                     | Hackathon: SIH 2026                      | Security Class: TACTICAL / SENSITIVE |
| Architecture: React 18 + FastAPI + SQLite | Vision Engine: YOLOv8 + ByteTrack + ANPR | Date of Dossier: September 2026      |

### 1. Executive Summary & Problem Formulation

Modern border control and perimeter defense rely heavily on conventional closed-circuit television (CCTV) cameras. However, human-monitored feeds suffer from high operator fatigue, cognitive overload, lack of real-time situational context, and delayed incident response times. **DRISHTIX** transforms traditional camera infrastructure into an autonomous, event-driven tactical command center that unifies multi-spectrum video analytics, mathematical geofencing, automatic license plate recognition (ANPR), radar GIS tracking, and automated Quick Response Force (QRF) dispatching into a single glass cockpit.

### 2. Full-Stack System Architecture

| Layer             | Technologies                                              | Key Responsibilities & Architectural Role                                                                                                                                                 |
|-------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Console  | React 18, Vite, Tailwind CSS, Lucide Icons, Web Audio API | Military HUD dark command center, real-time polling synchronizer, multi-tab switching, procedural 2D HTML5 Canvas rendering engine with optical/thermal/night-vision shaders.             |
| Backend API       | FastAPI (Python 3.11), Uvicorn ASGI Server, Pydantic v2   | High-concurrency asynchronous REST API, background worker threads, strict request/response validation, CORS middleware, and progressive tactical demo orchestrator.                       |
| Data Storage      | SQLite 3, SQLAlchemy ORM                                  | Relational schema for persistent management of Cameras, Events, Security Alerts, Virtual Geofences, ANPR Hotlists, QRF Dispatches, and System Parameters.                                 |
| Edge CV AI Engine | YOLOv8 + ByteTrack + ANPR OCR + Ray-Casting               | Real-time multi-class object detection (Pedestrians, Vehicles, Animals), trajectory tracking, polygon containment, directional tripwire intersection testing, and speed radar estimation. |

### 3. Tab-by-Tab In-Depth Operational Breakdown

The DRISHTIX command dashboard provides 8 specialized tactical modules accessible via the sidebar navigation:

#### TAB 1: Live Surveillance Matrix (CameraFeedGrid)

**Purpose:** Multi-channel synchronized tactical command matrix monitoring 4 live border surveillance nodes simultaneously.

- Camera Nodes:** CAM-01 (Border Patrol Alpha - PTZ), CAM-02 (Checkpoint Bravo - Entry Gate), CAM-03 (Main Gate - ANPR Optical), CAM-04 (Restricted Zone Delta - Thermal FLIR).
- Simulated Canvas Engine:** High-performance HTML5 2D Canvas procedurally rendering scanlines, optical noise, targeting reticles, and thermal false-color palettes.
- Multi-Spectrum Modes:** One-click switching between *Optical RGB*, *Night Vision (Green Phosphor Amplification)*, and *Thermal FLIR (White/Ironbow Heatmap)*.
- Live Telemetry & HUD:** Dynamic bounding boxes with confidence scores (e.g. Person 97%), PTZ angles, compass heading, and edge FPS counters.
- Interactive PTZ Controls:** Directional jog controls (Pan, Tilt, Zoom In/Out, Center Reset) allowing operators to steer surveillance angles in real time.

#### TAB 2: Tactical Radar GIS (TacticalMap)

**Purpose:** 2D Cartesian & Geospatial radar command view mapping coordinate nodes, camera field-of-view (FOV) coverage, and live target movement vectors.

- Tactical Grid:** Interactive radar coordinate system centered around Sector 7 (Lat: 32.7266° N, Lng: 74.8570° E).
- Camera FOV Visual Cones:** Geometric radar cones rendered at each node's exact heading angle and spread beam (e.g. 65° FOV at 45° azimuth).
- Dynamic Radar Blips & Tracks:** Displays real-time movement vectors for pedestrians, vehicles, and assigned QRF patrol units (Bravo-1, Alpha-2).
- Perimeter Geofence Polygons:** Visualized perimeter boundaries that pulse red upon intrusion, alerting operators to the exact breach coordinates.

#### TAB 3: Virtual Fence Studio (VirtualFenceStudio)

**Purpose:** Visual geofence editor enabling commanders to draw custom restricted zones and tripwires directly on live video feeds.

- Drawing Modes:** Supports multi-point closed Polygons (Restricted Areas / No-Go Zones) and linear Tripwires (Directional Entry/Exit Lines).
- Mathematical Intrusion Detection:** Uses ray-casting point-in-polygon algorithms and vector line-intersection algorithms to detect when a ByteTrack trajectory crosses a boundary.
- Directional & Risk Sensitivity:** Operators configure direction filters (*Inbound Only*, *Outbound Only*, or *Bidirectional*), sensitivity thresholds (1–100%), and severity levels (CRITICAL, HIGH, MEDIUM, LOW).
- Active Alarm State:** Instant perimeter highlighting and audio siren trigger when an unauthorized entity penetrates the perimeter.

TAB 4: ANPR Vehicle Hub (ANPRHub)

**Purpose:** Automated vehicle surveillance, optical license plate extraction, speed estimation radar, and national hotlist watchlist matching.

- **Optical OCR & Plate Parsing:** Automatically extracts vehicle license plates (e.g. MH01AB1234, DL08CX5678), state codes, and vehicle types (SUV, Truck, Sedan, Motorcycle).
- **Automated Hotlist / Watchlist Matching:** Compares every detected plate in real time against an encrypted database of stolen vehicles, contraband flags, and wanted suspects.
- **Instant Tactical Alarm:** Immediate high-priority red alert banner and audible warning tone whenever a watchlist plate is recognized.
- **Radar Speed Calculation:** Analyzes multi-frame bounding box displacement to estimate vehicle velocity and flag speed violations at entry checkpoints.
- **Plate Search & Watchlist Management:** Allows operators to search historical plate logs and register new suspect plates directly into the database.

TAB 5: Incident Command & Quick Response (IncidentCommander)

**Purpose:** Central dispatch console for triaging active security breaches, assigning Quick Response Force (QRF) units, and managing tactical SOPs.

- **Incident Triage Queue:** Active incidents sorted by threat severity (CRITICAL, HIGH, MEDIUM, LOW) with confidence scores and timestamps.
- **Standard Operating Procedure (SOP) Checklist:** Guided interactive checklist for operators: 1. *Verify Video Feed* → 2. *Trigger Sector Siren* → 3. *Dispatch QRF Unit* → 4. *Lock Checkpoint Gate* → 5. *Notify Border HQ*.
- **Patrol Unit Dispatch:** Assigns nearest response team (e.g. QRF Patrol Bravo-1), calculates estimated time of arrival (ETA), and tracks response status (DISPATCHED, EN\_ROUTE, ON\_SCENE, RESOLVED).
- **Incident Resolution Notes:** Records operator notes and formal debrief records into persistent database storage.

TAB 6: Evidence Vault & Forensic Archive (EvidenceVault)

**Purpose:** Tamper-evident digital forensic vault storing cryptographically verified incident snapshots and audit trails for legal and intelligence use.

- **Tamper-Evident SHA-256 Hashing:** Every captured visual snapshot and associated metadata string is hashed with SHA-256 to guarantee non-repudiation and chain of custody.
- **Metadata Stored:** Camera ID, Exact Timestamp, Detection Bounding Box Coordinates, Confidence %, Classification, Operator ID, and GPS Geo-tag.
- **Advanced Filtering:** Filter by Camera ID, Event Category, Risk Severity, Date Range, and Hash Verification.
- **Forensic Export Dossier:** Single-click export of structured forensic evidence packages suitable for intelligence reports and legal prosecution.

TAB 7: Threat Intelligence & Analytics (AnalyticsView)

**Purpose:** High-level situational analytics, risk heatmaps, temporal breach histograms, and Edge AI performance metrics.

- **Key Metric Indicators:** Total Events, Critical Incursions, Object Breakdown (Pedestrians vs Vehicles vs Animals), ANPR Scans, and False-Positive Rates.
- **Hourly Threat Distribution:** 24-hour histogram detailing peak intrusion hours and suspicious nocturnal movement clusters.
- **Sector Vulnerability Ranking:** Evaluates breach frequency by border sector to recommend patrol reallocations.
- **Edge AI Telemetry:** Monitors inference latency (average 14ms on GPU / 38ms on CPU), FPS throughput, and model memory footprint.

TAB 8: System Settings & Edge Config (SettingsView)

**Purpose:** System administration, camera RTSP stream configuration, AI model threshold tuning, integration webhooks, and storage retention.

- **Camera Node Management:** Configure RTSP/IP stream URIs, camera labels, frame rates, and GPS coordinates.
- **AI Hyperparameter Tuning:** Real-time sliders for YOLO confidence threshold (0.10–0.99), NMS IoU threshold, and ByteTrack track buffer length.
- **Notification Gateways:** Webhook triggers, MQTT broker integration for IoT sirens/gate locks, and SMS/Telegram emergency alerts.
- **Data Retention & Maintenance:** Automated log rollover policies, database backup/reset, and Role-Based Access Control (RBAC).

4. Tactical Progressive Demonstration Sequence

Clicking **START DEMO** in the top navigation bar triggers a progressive tactical simulation across all 4 border nodes:

| Timeline | Node / Camera             | Tactical Event Triggered                  | System Action & Tactical Response                                                                |
|----------|---------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------|
| T + 0s   | System Engine             | Demo Sequence Initiated                   | Spins up background CV simulation engine; audio chime sounds; status set to ACTIVE.              |
| T + 2s   | CAM-01 (Alpha Patrol)     | Pedestrian Movement Detected              | YOLOv8 bounding box acquired (Confidence 96%); green blip appears on radar map.                  |
| T + 7s   | CAM-02 (Checkpoint Bravo) | Vehicle Inbound Detected                  | Vehicle speed estimated at 42 km/h; logged into live surveillance event queue.                   |
| T + 14s  | CAM-03 (Gate ANPR)        | ANPR Watchlist Match (Hit)                | Plate <b>MH01AB1234</b> matched against Hotlist; HIGH risk warning tone sounds.                  |
| T + 22s  | CAM-04 (Restricted Delta) | <b>Virtual Fence Intrusion (CRITICAL)</b> | Ray-casting detects fence breach; <b>CRITICAL SIREN</b> sounds; automated QRF ticket dispatched. |

5. Core REST API Endpoint Reference

| HTTP Method | Endpoint              | Description & Return Payload                                                |
|-------------|-----------------------|-----------------------------------------------------------------------------|
| GET         | /api/health           | System health check, online edge node count, AI engine status.              |
| GET / POST  | /api/cameras          | Query active camera nodes, telemetry, PTZ presets, and stream URLs.         |
| POST        | /api/cameras/{id}/ptz | Issue PTZ jog commands (pan, tilt, zoom) to physical or simulated nodes.    |
| GET         | /api/events           | Fetch real-time event feed with risk ratings, bounding boxes, and metadata. |
| GET / POST  | /api/virtual-fences   | Manage drawn polygon coordinates, tripwires, sensitivity, and arming state. |
| GET / POST  | /api/watchlist        | Manage ANPR vehicle hotlist database for instant match triggering.          |
| GET / POST  | /api/dispatch         | Create and track Quick Response Force (QRF) patrol deployments.             |
| GET         | /api/analytics        | Aggregate metrics, hourly distribution histograms, and risk breakdowns.     |
| POST        | /api/start-demo       | Launch multi-stage progressive tactical demonstration sequence.             |
| POST        | /api/reset-demo       | Flush simulated events and reset state back to clean baseline.              |

6. Operational Impact & Conclusion

DRISHTIX provides a transformative, force-multiplying paradigm shift for border security and critical infrastructure defense. By fusing autonomous computer vision detection, mathematical geofencing, automatic license plate recognition, 2D radar GIS tracking, and structured incident response workflows into a single glass cockpit, DRISHTIX drastically reduces threat reaction times from minutes to seconds while preserving an immutable cryptographic chain of evidence.